A SHORT OBSTRUCTION TO UNIFORM PRIME-GAP ASYMPTOTICS AND THE ADDITIVE STRUCTURE OF THE ADMISSIBLE CONSTANTS

HRISHI SUNDER, SOURISH KUMRAWAT, AND KIREET CHERI

ABSTRACT. Let p_n be the n-th prime, let $g_n = p_{n+1} - p_n$, and put

$$d(x) = \max_{p_n < x} g_n.$$

For a fixed constant C, consider the uniform assertion

$$\pi(y + Cd(x)) - \pi(y) \sim \frac{Cd(x)}{\log y} \qquad (x/2 < y < x).$$

We give a compressed record-gap obstruction showing that two constants whose separation is less than 1 cannot both satisfy this assertion. We then prove that no $0 < C < 1$ can satisfy it by the same gap obstruction, while $C = 1$ is excluded by Westzynthius' theorem on large prime gaps. We then combine this with Tao's difference observation: if the assertion holds for C_1 and C_2, then it also holds for $C_1 - C_2$. Consequently the set of admissible constants is either trivial, or, under the natural signed convention, a cyclic lattice $C_0\mathbb{Z}$ for a unique $C_0 > 1$.

1. The uniform assertion

Let p_n denote the n-th prime and write

$$g_n = p_{n+1} - p_n.$$

For $x > 2$, define

$$d(x) = \max_{p_n < x}(p_{n+1} - p_n).$$

Thus a prime gap is counted as soon as its left endpoint is below x; its right endpoint need not be below x.

The motivating question is Erdős Problem #1138, as listed by Bloom [2]; its original source is [1, 1.3].

For fixed $C \neq 0$, let $A(C)$ denote the assertion

$$\sup_{x/2<y<x} \left| \frac{\pi(y + Cd(x)) - \pi(y)}{Cd(x)/\log y} - 1 \right| \longrightarrow 0 \qquad (x \to \infty). \tag{1}$$

For the structural statement, put

$$\mathcal{C} = \{0\} \cup \{C \neq 0 : A(C) \text{ holds}\},$$

with 0 adjoined only as the additive identity.

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We shall use three standard inputs. First, prime gaps are unbounded, so strict record prime gaps occur infinitely often. Second, by the prime number theorem,

$$d(x) = o(x). \tag{2}$$

Indeed $p_{n+1}/p_n \to 1$, so every prime gap whose left endpoint is below x is $o(x)$. Third, we shall use Westzynthius' theorem, namely

$$\limsup_{n\to\infty} \frac{p_{n+1} - p_n}{\log p_n} = \infty. \tag{3}$$

Equivalently, maximal prime gaps are not asymptotic to the average prime gap size $\log p$; in fact their normalized limsup is infinite [3].

2. The compressed record-gap obstruction

Theorem 2.1. *Let $1 < C_1 < C_2$ and suppose*

$$0 < C_2 - C_1 < 1.$$

Then $A(C_1)$ and $A(C_2)$ cannot both hold.

Proof. Choose strict record prime gaps

$$(P_k, Q_k) = (p_{n_k}, p_{n_k+1}), \qquad D_k = Q_k - P_k,$$

so that $D_k > g_m$ for all $m < n_k$. Put

$$x_k = P_k + 1.$$

For all large k, we have $P_k < x_k < Q_k$, and since (P_k, Q_k) is a strict record gap,

$$d(x_k) = D_k.$$

Let

$$\eta = C_2 - C_1 \in (0, 1), \qquad y_k = P_k - C_1 D_k.$$

By (2), we have $D_k = o(P_k)$, so for all large k,

$$x_k/2 < y_k < x_k.$$

Moreover

$$y_k + C_1 D_k = P_k, \qquad y_k + C_2 D_k = P_k + \eta D_k < Q_k.$$

Since there are no primes in (P_k, Q_k),

$$\pi(y_k + C_1 D_k) = \pi(P_k) = \pi(P_k + \eta D_k) = \pi(y_k + C_2 D_k).$$

Hence, setting

$$N_k = \pi(y_k + C_1 D_k) - \pi(y_k) = \pi(y_k + C_2 D_k) - \pi(y_k),$$

the two asserted asymptotics would give

$$N_k \sim \frac{C_1 D_k}{\log y_k} \qquad \text{and} \qquad N_k \sim \frac{C_2 D_k}{\log y_k}.$$

Thus $C_1 = C_2$, a contradiction. □

Corollary 2.2. *The assertion $A(C)$ cannot hold for every fixed $C > 1$.*

3. No constants in $(0,1]$

Theorem 3.1. *No constant $C \in (0,1]$ satisfies $A(C)$.*

Proof. Again let (P_k, Q_k) be strict record prime gaps and set

$$D_k = Q_k - P_k.$$

Choose, for instance,

$$x_k = P_k + \frac{1}{2}D_k.$$

Then $P_k < x_k < Q_k$, so $d(x_k) = D_k$. All starting points below are admissible for all large k, since $D_k = o(P_k)$.

First suppose $0 < C < 1$. Put

$$y_k = P_k + \frac{1-C}{2}D_k.$$

Then both y_k and

$$y_k + CD_k = P_k + \frac{1+C}{2}D_k$$

lie strictly inside the prime-free interval (P_k, Q_k). Hence

$$\pi(y_k + CD_k) - \pi(y_k) = 0,$$

whereas $A(C)$ would require the normalized ratio to tend to 1. Thus $A(C)$ fails for every $0 < C < 1$.

It remains to exclude $C = 1$. By Westzynthius' theorem, the record gaps may be chosen so that

$$\limsup_{k\to\infty} \frac{D_k}{\log P_k} = \infty.$$

Indeed, if some gap g_n has large ratio $g_n/\log p_n$, then the record gap up to that point has length at least g_n and left endpoint at most p_n, so its ratio is at least as large.

Now take

$$x_k = P_k + 1, \qquad y_k = P_k.$$

For all large k, $P_k < x_k < Q_k$, so again $d(x_k) = D_k$, and y_k is admissible. Since $Q_k = P_k + D_k$ is the next prime after P_k,

$$\pi(P_k + D_k) - \pi(P_k) = \pi(Q_k) - \pi(P_k) = 1.$$

Thus $A(1)$ would force

$$1 \sim \frac{D_k}{\log P_k}$$

along all record gaps, contradicting the preceding limsup. Hence $A(1)$ fails. □

4. The difference observation

The following observation is due to Terence Tao.

Lemma 4.1 (Difference closure)**.** *Suppose $0 < C_2 < C_1$. If $A(C_1)$ and $A(C_2)$ hold, then $A(C_1 - C_2)$ holds. Consequently, the positive admissible constants are closed under positive differences. Under the natural signed convention for (1), $\mathcal{C}$ is closed under subtraction.*

Proof. Put

$$\delta = C_1 - C_2, \qquad d = d(x).$$

Let $x/2 < y < x$. Since $d = o(x)$, all shifts of y by a fixed multiple of d have logarithm $(1 + o(1)) \log y$, uniformly in $x/2 < y < x$.

If $y + \delta d < x$, then both y and $y + \delta d$ are admissible starting points. Hence

$$\begin{aligned} \pi(y + \delta d) - \pi(y) &= \big[\pi(y + C_1 d) - \pi(y)\big] - \big[\pi(y + C_1 d) - \pi(y + \delta d)\big] \\ &\sim \frac{C_1 d}{\log y} - \frac{C_2 d}{\log(y + \delta d)} \sim \frac{\delta d}{\log y}. \end{aligned}$$

If $y + \delta d \geq x$, put

$$z = y - C_2 d.$$

Then for all large x,

$$x/2 < z < x.$$

Indeed $y \geq x - \delta d$, so $z \geq x - C_1 d > x/2$. Therefore

$$\begin{aligned} \pi(y + \delta d) - \pi(y) &= \big[\pi(z + C_1 d) - \pi(z)\big] - \big[\pi(z + C_2 d) - \pi(z)\big] \\ &\sim \frac{C_1 d}{\log z} - \frac{C_2 d}{\log z} \sim \frac{\delta d}{\log y}. \end{aligned}$$

The estimates are uniform in y, so $A(\delta)$ follows. □

Corollary 4.2. *No two distinct positive admissible constants can differ by at most* 1*.*

Proof. A difference in $(0, 1)$ is excluded by Theorem 2.1. A difference equal to 1 is excluded by Lemma 4.1 together with Theorem 3.1. □

5. Structure of the set of constants

Theorem 5.1. *Under the natural signed convention for $A(C)$, either*

$$\mathcal{C} = \{0\},$$

or there is a unique $C_0 > 1$ such that

$$\mathcal{C} = C_0\mathbb{Z} = \{C_0 k : k \in \mathbb{Z}\}.$$

Equivalently, the positive admissible constants are either empty or exactly

$$\{C_0, 2C_0, 3C_0, \ldots\}.$$

Proof. By Lemma 4.1, $\mathcal{C}$ is an additive subgroup of $\mathbb{R}$ under the signed convention. Let

$$\mathcal{C}_+ = \mathcal{C} \cap (0, \infty).$$

By Theorem 3.1,

$$\mathcal{C}_+ \cap (0, 1] = \emptyset.$$

If $\mathcal{C}_+ = \emptyset$, then $\mathcal{C} = \{0\}$. Otherwise set

$$C_0 = \inf \mathcal{C}_+.$$

Then $C_0 \geq 1$.

We first show that $C_0 \in \mathcal{C}_+$. If not, then there are distinct $a, b \in \mathcal{C}_+$ with

$$C_0 < a < b < C_0 + 1.$$

Since $\mathcal{C}$ is closed under subtraction, $b - a \in \mathcal{C}_+$. But $0 < b - a < 1$, contradicting Theorem 3.1. Hence $C_0 \in \mathcal{C}_+$. In particular $C_0 > 1$.

Now let $C \in \mathcal{C}_+$. Write

$$C = mC_0 + r, \qquad m \in \mathbb{Z}_{\geq 0}, \qquad 0 \leq r < C_0.$$

Since $\mathcal{C}$ is a subgroup and $C, C_0 \in \mathcal{C}$, the remainder $r = C - mC_0$ lies in $\mathcal{C}$. By the minimality of C_0, this forces $r = 0$. Thus every positive element of $\mathcal{C}$ is an integer multiple of C_0. Conversely, because $\mathcal{C}$ is a subgroup and $C_0 \in \mathcal{C}$, every integer multiple of C_0 lies in $\mathcal{C}$. Therefore $\mathcal{C} = C_0\mathbb{Z}$. The generator is unique because it is the least positive element of $\mathcal{C}$. □

6. Conclusion

The obstruction is caused by the definition of $d(x)$: a record prime gap is counted as soon as its left endpoint is below x, even if the gap extends far past x. Placing x just inside such a gap and taking $y = P - C_1 D$ makes the two endpoints

$$y + C_1 D = P, \qquad y + C_2 D = P + (C_2 - C_1)D$$

land in the same prime-free gap whenever $0 < C_2 - C_1 < 1$. The two prime counts are then identical, but the predicted main terms differ by the factors C_1 and C_2. Together with the exclusion of $(0, 1]$ and Tao's difference observation, this forces the admissible constants to be either trivial or a single arithmetic lattice.

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References

[1] Various, *Some of Paul's favorite problems*. Booklet produced for the conference "Paul Erdős and his mathematics", Budapest, July 1999 (1999).

[2] T. F. Bloom, *Erdős Problem #1138*, https://www.erdosproblems.com/1138, accessed 2026-05-22.

[3] E. Westzynthius, Über die Verteilung der Zahlen, die zu den n ersten Primzahlen teilerfremd sind, *Commentationes Physico-Mathematicae* **5** (1931), 1–37.